

$$\nabla^2=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2}{\partial\varphi^2}+\frac{\partial^2}{\partial z^2}$$

$u=U(r,\varphi)Z(z)$ $U=R(r)\Phi(\varphi)$

$$\begin{cases}Z''+\lambda Z=0\\ \Phi''+\mu\Phi=0 \xrightarrow{\text{周期条件}} \mu=m^2\\ \frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right)+\left(k^2-\lambda-\frac{m^2}{r^2}\right)R=0\end{cases}$$

(Bessel方程)

$Z''-\lambda Z=0$

分离变量 柱坐标

$\nabla^2u+k^2u=0$

(Helmholtz方程)

$\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right)+\left(k^2-\frac{m^2}{r^2}\right)R=0$ 齐次化 $\lambda=0$

$x=kr$ $x^2R''+xR'+(x^2-m^2)R=0$ → $R_m=C_mJ_m(kr)+D_mN_m(kr)$

(m阶Bessel方程)

Laplace方程 $k=0$

$\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right)-(\lambda+\frac{m^2}{r^2})R=0$ 齐次化 $m=0$

$\begin{cases}Z''-\lambda Z=0\\ rR''+R'+\lambda rR=0\end{cases}$

$Z=A\sinh\sqrt{\lambda}z+B\cosh\sqrt{\lambda}z$ $Z(0)=0$ → $Z=A\sinh\sqrt{\lambda}z$

$R(a)=0$ $\lambda_i=\left(\frac{\mu_i}{a}\right)^2$, $i=1,2,\cdots$

$R_i(r)=J_0(\frac{\mu_i}{a}r)$

$x=\sqrt{\lambda}r$ $x^2R''+xR'+x^2R=0$ → $R(r)=J_0(\sqrt{\lambda}r)$

(0阶Bessel方程)

$Z(0)=Z(0)=0$

$u|_{r=a}=f(\phi,z)$

$\frac{d}{dr}\left(r\frac{dR}{dr}\right)-(\lambda r+\frac{m^2}{r})R=0$

$\begin{cases}\lambda_n=\left(\frac{n\pi}{h}\right)^2\\ Z_n=\sin\frac{n\pi}{h}z\end{cases}$ $x=\frac{n\pi}{h}r$ $x^2R''+xR'+(-x^2-m^2)R=0$ $t=ix$ → $t^2R''+tR'+(t^2-m^2)R=0$ → $R=CJ_m(ix)+DN_m(ix)$

(虚宗量m阶Bessel方程)

$R_m=I_m\left(\frac{n\pi r}{h}\right)$

$R(a)=0$ $I_m(x)$ $K_m(x)$ $0=0$

$x^2R''+2xR'+[x^2-l(l+1)]R=0$ $R(x)=\frac{\sqrt{\pi}}{2x}v(x)$ → $x^2v''+xv'+[x^2-(l+\frac{1}{2})^2]v=0$

(球Bessel方程) (l+1/2阶Bessel方程)

→ $v_l(x)=C_lJ_{l+\frac{1}{2}}(x)+D_lN_{l+\frac{1}{2}}(x)$

$j_l(x)=\frac{\sqrt{\pi}}{2x}J_{l+\frac{1}{2}}(x)$ $n_l(x)=\frac{\sqrt{\pi}}{2x}N_{l+\frac{1}{2}}(x)$

$R=C_lj_l(x)+D_ln_l(x)$

$R(0)<\infty$ $D_l=0$

$R_l=j_l(kr)$

$R(a)=0$ $k_{mn}=\frac{\mu_n^{(l)}}{a}$

本征函数 $R_{mn}(r)=j_l(\frac{\mu_n^{(l)}}{a}r)$

Helmholtz方程 球内问题

$u(r,\theta,\phi)=\sum_{n=1}^{\infty}\sum_{l=0}^{\infty}\sum_{m=0}^{l}j_l(\frac{\mu_n^{(l)}}{a}r)(A_{lm}S_{lm}+B_{lm}S_{lm})$

球坐标 分离变量 $u=R(r)S(\theta,\phi)$

$\begin{cases}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\left(k^2-\frac{\lambda}{r^2}\right)R=0\\ \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial S}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2S}{\partial\varphi^2}+\lambda S=0\end{cases}$

$k\neq0$ $x=kr$ $\gamma=l(l+1)$

$r^2R''+2rR'-\lambda R=0$ (欧拉方程)

$R_l=C_lr^l+\frac{D_l}{r^{l+1}}$ $R(0)<\infty$ → $R_l=C_lr^l$

$S=\Theta(\theta)\Phi(\phi)$

$\begin{cases}\Phi''+\mu\Phi=0 \xrightarrow{\text{周期条件}} \mu=m^2\\ \frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+\left(\lambda-\frac{m^2}{\sin^2\theta}\right)\Theta=0\end{cases}$

$\begin{cases}x=\cos\theta\\ y=\Theta\end{cases}$ → $(1-x^2)y''-2xy'+\left(\lambda-\frac{m^2}{1-x^2}\right)y=0$

齐次边界 $m=0$

$(1-x^2)y''-2xy'+\lambda y=0$ (Legendre)

本征值 $\lambda=l(l+1)$, $l=0,1,\cdots$

本征函数 $P_l(x)$ $x=\cos\theta$ → $P_l(\cos\theta)$

权函数1 权函数 $\sin\theta$

$m=1,2,\cdots$

$(1-x^2)y''-2xy'+\left(\lambda-\frac{m^2}{1-x^2}\right)y=0$

(m阶连带Legendre)

本征值 $\lambda=l(l+1)$, $l=m,m+1,\cdots$

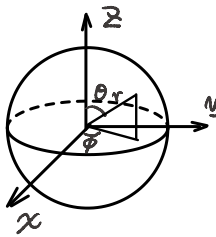
本征函数 $P_l^m(x)=(-1)^m(1-x^2)^{\frac{m}{2}}[P_l(x)]^{(m)}$ $x=\cos\theta$ → $P_l^m(\cos\theta)$

权函数1 权函数 $\sin\theta$

球谱函数 $\begin{cases}S_{lm1}=P_l^m(\cos\theta)\cos m\varphi, m=0,1,\cdots,l\\ S_{lm2}=P_l^m(\cos\theta)\sin m\varphi, m=1,2,\cdots,l\end{cases}$

Laplace方程 球内问题

$u(r,\theta,\phi)=\sum_{l=0}^{\infty}\sum_{m=0}^{l}r^l(A_{lm}S_{lm}+B_{lm}S_{lm})$



$$\nabla^2=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$

圆柱边界为0的 Helmholtz方程 圆内问题

本征函数 $\begin{cases}J_m(\frac{\mu_i^{(m)}}{a}r)\cos m\varphi, m=0,1,\cdots\\ J_m(\frac{\mu_i^{(m)}}{a}r)\sin m\varphi, m=1,2,\cdots\end{cases}$ → $u=\sum_{m=1}^{\infty}\sum_{i=1}^{\infty}J_m(\frac{\mu_i^{(m)}}{a}r)(A_{mi}\cos m\varphi+B_{mi}\sin m\varphi)$

$R_{mi}(r)=J_m(\frac{\mu_i^{(m)}}{a}r)$

$R(a)=0$ $k_{mi}=\frac{\mu_i^{(m)}}{a}$, $i=1,2,3\cdots$

$R_m=C_mJ_m(kr)$

$R(0)<\infty$ $D_m=0$

柱面边界为0的Laplace方程 柱内问题

$u=\sum_{i=1}^{\infty}C_i\sinh\left(\frac{\mu_i}{a}z\right)J_0(\frac{\mu_i}{a}r)$

本征函数 $\sinh(\frac{\mu_i}{a}z)J_0(\frac{\mu_i}{a}r)$

$Z_i=\sinh\frac{\mu_i}{a}z$

$\int_0^a f(r)\gamma_0r^2dr$

上底面边界为0的 Laplace方程 柱内问题

$u=\sum_{m=0}^{\infty}\sum_{n=1}^{\infty}(A_{mn}\cos m\varphi+B_{mn}\sin m\varphi)I_m(\frac{n\pi}{h}r)\sin(\frac{n\pi}{h}z)$

本征函数的模方

$$\|P_l(x)\|^2=\frac{2}{2l+1}\qquad \|P_l^m(x)\|^2=\frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}$$

$$\|S_{lm1}\|^2=\frac{(l+m)!}{(l-m)!}\frac{2\pi}{2l+1}(1+\delta_{m0})\qquad \|S_{lm2}\|^2=\frac{(l+m)!}{(l-m)!}\frac{2\pi}{2l+1}$$

$$\|J_m(\frac{\mu_i^{(m)}}{a}r)\|^2=\frac{a^2}{2}[J_{m+1}(\mu_i^{(m)})]^2$$

递推关系

$$(2l+1)xP_l(x)=(l+1)P_{l+1}(x)+lP_{l-1}(x)\qquad P'_{l+1}(x)+P'_{l-1}(x)=2xP'_l(x)+P_l(x)$$

$$\frac{d}{dx}[x^\nu J_\nu(x)]=x^\nu J_{\nu-1}(x)\qquad \frac{d}{dx}[x^{-\nu}J_\nu(x)]=-x^{-\nu}J_{\nu+1}(x)$$

积分公式

$$\int_{-1}^1 x^k P_l(x)\,dx=\begin{cases}0,\ k<l\text{ 或 }k,l\text{ 奇偶性不同}\\ \frac{1}{2^l}C_{l+2n}^{l}B(n+\frac{1}{2},l+1)\\ =2^{l+1}\frac{(l+2n)!(l+n)!}{n!(2l+2n+1)!},\ k=l+2n\end{cases}$$

$$B(p,q)=\int_0^1 t^{p-1}(1-t)^{q-1}dt=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

$$\Gamma(x)=\int_0^{+\infty}t^{x+1}e^{-t}dt$$

$$\int_0^1 P_{2n+1}(x)\,dx=(-1)^n\frac{(2n)!}{(2^n n!)^2(2n+2)}\qquad \int_0^1 xP_{2n}(x)\,dx=(-1)^{n+1}\frac{(2n)!}{(2^n n!)^2(2n-1)(2n+2)}$$

$$\sum_{l=0}^{\infty}P_l(x)t^l=\frac{1}{\sqrt{1-2xt+t^2}},\ |t|\le 1$$

$$\int_0^1(1-x^2)J_0(\mu x)x\,dx=\frac{2}{\mu^2}J_2(\mu)=\frac{4}{\mu^3}J_1(\mu)\qquad \int_0^1 J_0(\mu x)x\,dx=\frac{J_1(\mu)}{\mu}$$

$$\|P_l\|^2=\frac{2}{2l+1}\qquad \|S_{lm1}\|^2=\frac{(l+m)!}{(l-m)!}\frac{2}{2l+1}(1+\delta_{m0})\pi$$

$$\|S_{lm2}\|^2=\frac{(l+m)!}{(l-m)!}\frac{2}{2l+1}\pi$$

$$P_0=1\quad P_1=x\quad P_2=\frac{1}{2}(3x^2-1)$$

$$P_1'=-(1-x^2)^{\frac{1}{2}}\quad P_2'=-3x(1-x^2)^{\frac{1}{2}}$$

$$P_2^2=3(1-x^2)$$

$$P_1'(\cos\theta)=-\sin\theta\qquad P_2'(\cos\theta)=-3\cos\theta\sin\theta$$

$$P_2^2(\cos\theta)=3\sin^2\theta$$